

EMERSON EIMS BUILDING SUITE PRO

Sample Residential Project

Location: Nairobi, Kenya
Total Area: 100 m² | Units: 1 | Storeys: 1
Architectural Style: Modern
Date: 27 April 2026

PROJECT INFORMATION	
Project ID	1367cb00-0338-4469-a0cf-558664a72f3d
Location	Nairobi, Kenya
Total Area	100 m²
Units / Apartments	1
Storeys	1
Bedrooms per Unit	1
Foundation Type	Pad
Style	Modern
Report Generated	2026-04-27 02:42

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VALIDATION ENGINE — AUDIT TRAIL

■ 1 BLOCKER(S) — DO NOT SUBMIT

- Kenya statutory IDs missing: plot_no, architect, engineer, qs. Required for BORAQS / EBK / NCA / county submission.

1 auto-correction(s) applied with code citation:

Field	Original	Auto-corrected to	Reason / Code citation
Foundation	(unspecified)	pad	Single-storey ≤ 120 m² → pad (BS 8004 §7.3)

Compliance Status: **BLOCKED — DO NOT SUBMIT**

1. DESIGN BRIEF & DESIGN CRITERIA

Item	Value
Project name	Sample Residential Project
Location	Nairobi, Kenya
Total floor area	100.0 m²
No. of storeys / floor count	1 (single source of truth)
Dwelling units	1
Bedrooms / unit	1
Architectural style	Modern
Foundation	PAD
Currency	KES
Plot / LR No.	— REQUIRED —
Coordinates (lat,lng)	-1.2921,36.8219
Title Deed	—
Architect (BORAQS)	— REQUIRED —
Structural Engineer (EBK)	— REQUIRED —
Quantity Surveyor (BORAQS)	— REQUIRED —
Client	—

Design loads (BS EN 1991-1-1 Cat A — residential):

Action	Value (kN/m²)	Reference
Permanent action gk (slab + finishes + partitions)	7.00	EC1 Table 6.1
Variable action qk	2.00	EC1 Table 6.2 Cat A
Design ULS w = 1.35 gk + 1.5 qk	12.45	EC0 §6.4.3.2
Design SLS w = gk + qk	9.00	EC0 §6.5.3

2. STRUCTURAL DESIGN CALCULATIONS

Calculations to BS EN 1992-1-1 (EC2) using $f_{ck} = 30$ MPa, $f_{yk} = 500$ MPa, cover per exposure class XC1 internal / XC3 external. All members hand-verified against simplified Eurocode 2 design rules; final detailing to be reviewed by the Structural Engineer of Record (EBK-registered) before issue for construction.

2.1 Slab design (one/two-way RC slab)

Parameter	Value
Short span L_x	9.13 m
Long span L_y	10.95 m
Behaviour	Two-way
Slab thickness h	350 mm (span/depth = 26)
Effective depth d	319.0 mm
Cover	25 mm (XC1)
Design moment M	64.33 kN·m / m
As required	488.0 mm ² / m
As minimum (EC2 §9.3.1.1)	455.0 mm ² / m
As provided	488.0 mm ² / m
Reinforcement	T12 @ 225 c/c BW

2.2 Beam design (simply-supported RC beam)

Parameter	Value
Clear span L	5.48 m
Beam section $b \times h$	225 × 450 mm
Effective depth d	404.0 mm
Cover	30 mm
Service load w	59.36 kN/m
Design moment M	222.6 kN·m
Design shear V	162.6 kN
As required (tension)	1333.0 mm ²
Main steel	7 no. T16 bot · 4 no. T16 top
Shear links	T8 @ 300 c/c

2.3 Column design (axially loaded RC column)

Parameter	Value
Axial load N (ULS)	316.0 kN
Section	225 × 225 mm
Concrete grade	C25/30 ($f_{ck} = 30$ MPa)
Steel grade	$f_{yk} = 500$ MPa
Steel ratio ρ	0.020 (2.0 %)
As required	1012.0 mm ²
Main steel	6 no. T16
Links	T8 @ 192 c/c

2.4 Foundation design (per BS 8004 / EN 1997-1)

Parameter	Value
Foundation type	PAD
Plan size	1.5 × 1.5
Thickness	750 mm
Concrete grade	C25/30
Cover	75 mm (XC2 ground contact)
Safe bearing pressure (presumed)	150 kPa
Column load to foundation	316.0 kN
Reinforcement	T12 @ 150 c/c BW bottom

■ The bearing pressure shown is a BS 8004 Table 1 *presumed* value pending a site investigation. A signed Geotechnical Investigation Report (boreholes, SPT, allowable bearing capacity) must be obtained from a geotechnical engineer before construction. EIMS records this gap honestly rather than fabricating SPT values.

3. BAR BENDING SCHEDULE (BS 8666)

Schedule of all reinforcement bars with mark, diameter, shape code, length per bar, number off, total length and weight. Steel grade 500B. Bend radii and bar shapes comply with BS 8666:2020 Table 2. Quantities to be verified by the Engineer before cutting and bending.

Bar Mark	Dia.	Shape Code	Length per bar (m)	No. off	Total length (m)	Weight (kg)
S1	T12	00	10.30	80	824.00	731.7
B1-bot	T16	00	5.98	48	286.91	453.0
B1-top	T16	37	5.98	16	95.64	151.0
B1-link	T8	51	0.85	144	122.40	48.3
C1	T16	00	3.50	24	84.00	132.6
C1-link	T8	51	0.60	72	43.20	17.1
F1	T12	00	3.00	48	144.00	127.9
					TOTAL	1661.6

Total reinforcement steel: 1,662 kg $\approx$ 1.66 tonnes

4. MEP DESIGN — ELECTRICAL, PLUMBING & DRAINAGE

Electrical design per IEC 60364 / KS IEC 60364. Plumbing & sanitary drainage per BS EN 12056-2 and Kenya NCC Building Code SEHS. Septic tank to NEMA EIA Regulations 2003.

4.1 Electrical load schedule

Load description	Value
Lighting (LED) @ 5 W/m ²	500 W
Small power sockets @ 25 W/m ²	2500 W
Cooker / hob (per unit)	6000 W
Water heaters	3000 W
Air conditioning (split units)	1500 W
TOTAL CONNECTED LOAD	13,500 W
Diversity factor	0.65
DESIGN MAXIMUM DEMAND	8,775 W (9.2 kVA)
Main breaker (single phase 230 V)	45 A

4.2 Final-circuit schedule

Circuit	Load	Cable	Protection	RCD
L1-Ln (Lighting)	500 W	1.5 mm ² T+E	10 A B-curve MCB	—
P1-Pn (Sockets)	2500 W	2.5 mm ² T+E	20 A B-curve MCB	30 mA RCD
C (Cooker)	6000 W	6 mm ² T+E	32 A B-curve MCB	—
W (1× water heater)	3000 W	4 mm ² T+E	25 A B-curve MCB	30 mA RCD
A1-A1 (AC splits)	1500 W	2.5 mm ² T+E	16 A C-curve MCB	—

4.3 Single-line diagram (text representation)

KPLC supply ■ kWh meter ■ Main isolator 45 A ■ Main DB (Type-B SPD)
■ Lighting circuits L1...Ln (1.5 mm² · 10 A)
■ Socket circuits P1...Pn (2.5 mm² · 20 A · 30 mA RCD)
■ Cooker C (6 mm² · 32 A)
■ Water heaters W (4 mm² · 25 A · 30 mA RCD)
■ AC splits A1...An (2.5 mm² · 16 A)
■ Earth busbar ■ copper earth-rod ≤ 1 Ω

4.4 Plumbing fixture schedule

Fixture	Qty
Water Closet (WC)	2
Wash-Hand Basin	2
Shower / Bath	2
Kitchen Sink	1
Laundry Sink	1
External Bib Tap	2

4.5 Drainage gradient table (BS EN 12056-2)

Pipe	Slope range	Min fall
100 mm soil pipe (WC)	1:40 to 1:80	≥ 1.00 m fall per 1 m
75 mm waste	1:40	0.025 m / m

50 mm waste	1:40	0.025 m / m
32 mm WHB / sink	1:20	0.050 m / m

4.6 Septic tank & soak-pit sizing

Parameter	Value
Design occupancy	2 persons (BS 8233 — bedrooms × 2)
Daily wastewater flow (180 L/p/d)	360 L/day
Septic tank volume (3-day retention)	1.08 m ³
Suggested tank dimensions	2.5 × 1.2 × 0.36 m (L×W×D)
Soak-pit area (porous soil, 30 L/m ² /d)	12.0 m ²
Soak-pit diameter (circular)	3.91 m

■ Final septic-system geometry must be confirmed by an on-site percolation test. A NEMA Environmental Impact Assessment (EIA) project report is required for systems > 50 persons or sites near surface watercourses.

5. BILL OF QUANTITIES (KQS-2025)

Contractor-grade BOQ in KES, structured by trade per the Kenya Quantity Surveyors Standard Method of Measurement. Rates are KQS-2025 indicative urban rates; tenderers shall price all items. All quantities to be verified on site.

A — PRELIMINARIES

Description	Unit	Qty	Rate	Amount
Site mobilisation, hoarding, signage	item	1.00	KSh 180,000	KSh 180,000
Insurance & NCA Levy	item	1.00	KSh 220,000	KSh 220,000
Site office + storage	item	1.00	KSh 95,000	KSh 95,000
			Trade subtotal	KSh 495,000

B — SUBSTRUCTURE

Description	Unit	Qty	Rate	Amount
Site clearance & strip topsoil	m²	130.00	KSh 120	KSh 15,600
Bulk excavation foundation trenches	m³	30.00	KSh 650	KSh 19,500
Hardcore filling 200 mm	m²	100.00	KSh 850	KSh 85,000
Lean concrete blinding 50 mm	m³	5.00	KSh 14,000	KSh 70,000
RC foundation C25/30	m³	30.00	KSh 26,000	KSh 780,000
			Trade subtotal	KSh 970,100

C — SUPERSTRUCTURE

Description	Unit	Qty	Rate	Amount
RC columns C25/30	m³	0.71	KSh 32,500	KSh 23,034
RC beams C25/30	m³	4.44	KSh 30,500	KSh 135,315
RC suspended slabs C25/30	m³	35.00	KSh 28,500	KSh 997,500
Reinforcement bars (BBS)	kg	1,661.60	KSh 165	KSh 274,164
Formwork to soffits & sides	m²	240.87	KSh 1,450	KSh 349,264
200 mm machine-cut block walling	m²	120.00	KSh 1,800	KSh 216,000
Plaster both sides	m²	240.00	KSh 1,100	KSh 264,000
			Trade subtotal	KSh 2,259,277

D — ROOFING

Description	Unit	Qty	Rate	Amount
Timber roof trusses & purlins	m²	100.00	KSh 2,200	KSh 220,000
0.6 mm box-profile coloured iron sheet	m²	105.00	KSh 1,450	KSh 152,250
9 mm gypsum ceiling on softwood brandering	m²	100.00	KSh 1,850	KSh 185,000
			Trade subtotal	KSh 557,250

E — FINISHES

Description	Unit	Qty	Rate	Amount
Floor ceramic / porcelain tiling	m²	85.00	KSh 2,400	KSh 204,000
Wall tiling (wet areas)	m²	18.00	KSh 2,200	KSh 39,600
Internal paint (Crown / Plascon emulsion)	m²	192.00	KSh 520	KSh 99,840
External paint (weather-shield)	m²	96.00	KSh 720	KSh 69,120

			Trade subtotal	KSh 412,560
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F — JOINERY

Description	Unit	Qty	Rate	Amount
Internal flush doors with hardware	each	4.00	KSh 18,000	KSh 72,000
External main doors hardwood	each	2.00	KSh 42,000	KSh 84,000
Aluminium-framed sliding windows	m²	15.00	KSh 9,500	KSh 142,500
			Trade subtotal	KSh 298,500

G — ELECTRICAL

Description	Unit	Qty	Rate	Amount
Mains supply, DB, MCB, RCDs (per IEC 60364)	item	1.00	KSh 165,000	KSh 165,000
Lighting points & circuits (~500 W)	point	18.00	KSh 2,200	KSh 39,600
Socket outlets (~2500 W)	point	12.00	KSh 2,400	KSh 28,800
			Trade subtotal	KSh 233,400

H — PLUMBING

Description	Unit	Qty	Rate	Amount
Cold + hot water reticulation, all fixtures	item	1.00	KSh 285,000	KSh 285,000
Septic tank + soak-pit (1.08 m³ + Ø3.91 m)	item	1.00	KSh 195,000	KSh 195,000
Stormwater drainage & rain-water disposal	item	1.00	KSh 145,000	KSh 145,000
			Trade subtotal	KSh 625,000

Cost summary

Item	Amount
A — PRELIMINARIES	KSh 495,000
B — SUBSTRUCTURE	KSh 970,100
C — SUPERSTRUCTURE	KSh 2,259,277
D — ROOFING	KSh 557,250
E — FINISHES	KSh 412,560
F — JOINERY	KSh 298,500
G — ELECTRICAL	KSh 233,400
H — PLUMBING	KSh 625,000
Subtotal of works	KSh 5,851,087
Contingency 7.5 %	KSh 438,832
Professional fees 8 % (Architect + Engineer + QS)	KSh 468,087
Sub-total before VAT	KSh 6,758,006
VAT 16 %	KSh 1,081,281
GRAND TOTAL	KSh 7,839,287
Cost per m² (KES)	KSh 78,393 / m²

6. STATUTORY COMPLIANCE PACK

All approvals required under Kenyan law before construction. The contractor and professional team shall ensure each item is closed prior to mobilisation.

Approval	Authority / Statute	Documents required
1. County Planning Approval	County Govt (Building Plans Section)	Stamped architectural & structural drawings, location plan, ownership docs, NCA
2. NCA Project Registration	National Construction Authority Act 2011	NCA-1 form, contractor registration certificate, levy 0.5 % of contract sum.
3. NEMA Project Report (PR)	EMCA 1999, EIA Regs 2003	Required for projects > 5 ha or near watercourses; PR submitted to NEMA county
4. Public Health Approval	Public Health Act Cap. 242	Septic system & water supply review by County Public Health Office.
5. EPRA / KPLC Energy Connection	EPRA Regulations	Electrical drawings & wayleave application; final inspection by EPRA-registered C
6. Fire Safety Certificate	County Fire Marshal	Means-of-escape, extinguisher schedule, hydrant layout (NFPA 241 / Kenya Fire
7. DOSHS Notification	OSHA 2007 §41	Notification before commencement; site safety plan, induction registers, ISO 4500

6.1 Professional certification (signature pages)

Role	Name & Reg.	Signature & Stamp	Date
Architect (BORAQS)	— REQUIRED —	_____	Date: _____
Structural Engineer (EBK)	— REQUIRED —	_____	Date: _____
Quantity Surveyor (BORAQS)	— REQUIRED —	_____	Date: _____
Site Manager (NCA)	— to be appointed —	_____	Date: _____
Client	—	_____	Date: _____

I certify that the design has been carried out in accordance with the relevant Kenyan codes (NCA Act 2011, EMCA 1999, OSHA 2007) and applicable Eurocodes / British Standards, and that I take professional responsibility for the section of work above corresponding to my registration.

8. PROJECT SUMMARY

Phase phase_1: Site Analysis

water_table_depth	3.5
water_table_source	regional_estimate
water_table_note	Regional estimate from climate/soil model -- verify with on-site geotechnical su
soil_type	clay
vegetation_coverage	0.65
slope_angle	8.25
suitability_pct	90
nearby_structures	survey_required
land_use	residential
climate_zone	tropical
analysis_method	regional_data_model
data_source	EIMS Regional Database + GPS Coordinates

Phase phase_2: Geotechnical Design

bearing_capacity_kPa	150
soil_classification	CH
recommended_foundation	STRIP
settlement_mm	40
slope_stability	stable
water_table_depth	3.5
excavation_depth	2.8

Phase phase_3: Foundation Design

foundation_type	PAD
foundation_depth_m	1.0
footing_width_m	1.29
total_building_load_kN	750.0
bearing_capacity_kPa	150
safety_factor	3.0
concrete_volume_m3	15.0
reinforcement_steel_kg	1350.0
soil_type	clay
water_table_factor	standard
design_standard	BS 8004 / Eurocode 7
bim_ready	True

Phase phase_4: Floor Plans

floor_count	1
total_area_sqm	100.0
bathrooms	1
rooms	{'bedrooms': 1, 'bathrooms': 1, 'living_areas': 1, 'kitchen': 1, 'extra_rooms':

style	modern
area_per_unit	100.0
building_code_compliance	True
area_efficiency_percent	72.5
cost_per_sqm_estimate	calculated_in_costing_phase

Phase phase_5: Electrical Design

total_connected_load_kW	8.06
max_demand_kW	4.84
diversity_factor	0.6
supply_type	single-phase 230V
main_current_A	21.0
lighting_points	6
socket_outlets	6
total_circuits	5
lighting_circuits	1
socket_circuits	1
fixed_appliance_circuits	3
distribution_boards	1

Phase phase_6: Plumbing Design

total_occupants	1
daily_demand_liters	150
peak_hourly_liters	15
total_fixture_units	13.5
main_supply_pipe_mm	25
cold_water_pipes	5
hot_water_pipes	3
waste_outlets	4
soil_stacks	1
hot_water_storage_liters	40
water_heater_kW	1.6
storage_tank_liters	1000

Phase phase_7: BOQ

material_items	77
total_material_cost	38036.27
currency	USD
items_by_category	{'external': 12, 'finishes': 18, 'plumbing': 14, 'electrical': 11, 'mep_hvac': 5
items	[{'item_no': 1, 'description': 'Portland Cement (50kg)', 'unit': 'bag', 'quantit
calculation_basis	{'total_area_m2': 100.0, 'stories': 1, 'units': 1, 'bedrooms_per_unit': 1, 'foot
waste_factor_applied	True
method	quantity_takeoff_from_dimensions

Phase phase_8: Infrastructure

solar_system	{'panels_kw': 2.0, 'panels_count': 5, 'battery_kWh': 3.0, 'estimated_cost_usd':
borehole_system	{'depth_meters': 8, 'estimated_yield_lpm': 15, 'pump_kw': 3, 'estimated_cost_usd
grid_option	{'connection_cost_usd': 200, 'monthly_tariff_estimate_usd': 15}
recommendation	grid_primary
estimated_annual_saving_usd	372.6

Phase phase_9: Landscape Design

lawn_area_m2	35.0
hardscape_area_m2	10.0
garden_area_m2	15.0
plants_count	120
irrigation_system	False
recommended_style	tropical_lush
estimated_cost_usd	345
maintenance_monthly_usd	6

Phase phase_10: Permits & Compliance

jurisdiction	Kenya
building_code	Kenya Building Code 2020
building_code_compliance	REQUIRES_VERIFICATION
permits_required	6
estimated_timeline_days	45
estimated_fees_usd	80
setback_compliance	{'front_m': 5, 'side_m': 2, 'rear_m': 3}
required_documents	['Architectural drawings (stamped)', 'Structural calculations (engineer-certifie

Phase phase_11: 3D Visualization

status	ready
model_endpoint	/api/drawings/3d-model

Phase phase_12: Live Costing

material_cost	4944715.1
labor_cost	556280.45
equipment_cost	395577.21
overhead	440079.64
contingency	633665.24
subtotal	6970317.64
contractor_profit	1045547.65
total_project_cost	8015865.29
cost_per_sqm	80158.65
labor_multiplier	0.25
region	Kenya
currency	KES

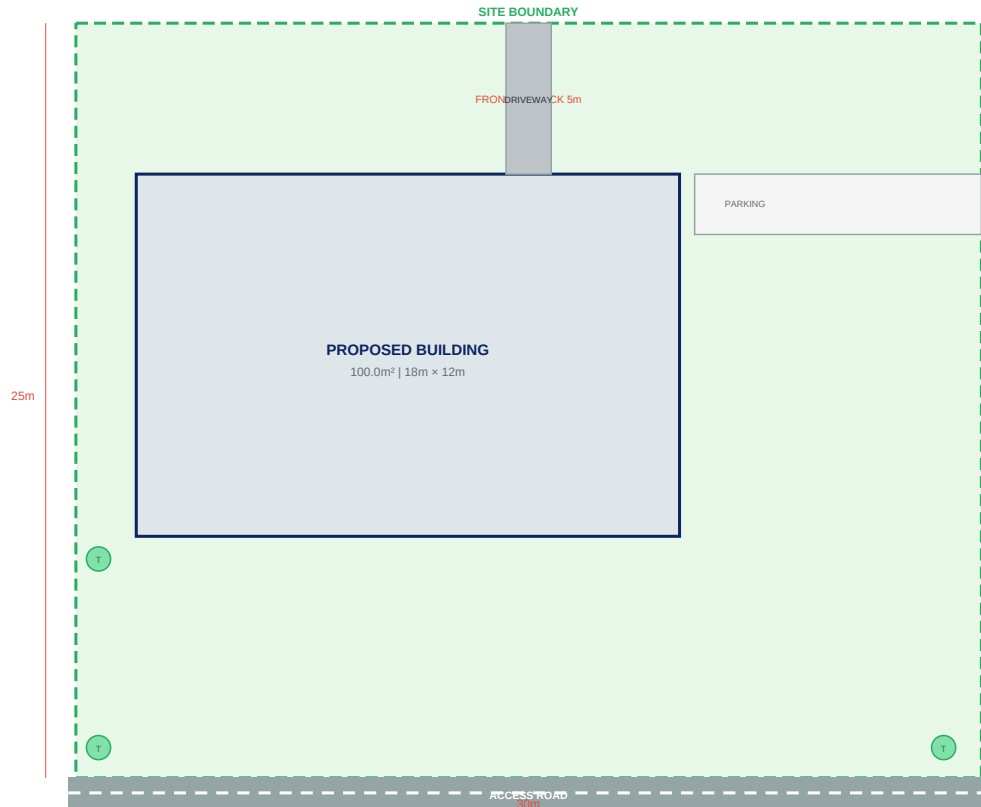
Phase phase_13: Integration Layer

status	complete
export_formats	['IFC2x3', 'FBX', 'DXF', 'PDF', 'XLSX']

9. SITE PLAN

Site layout showing building footprint, access roads, setbacks and landscaping. Area: 100 m²

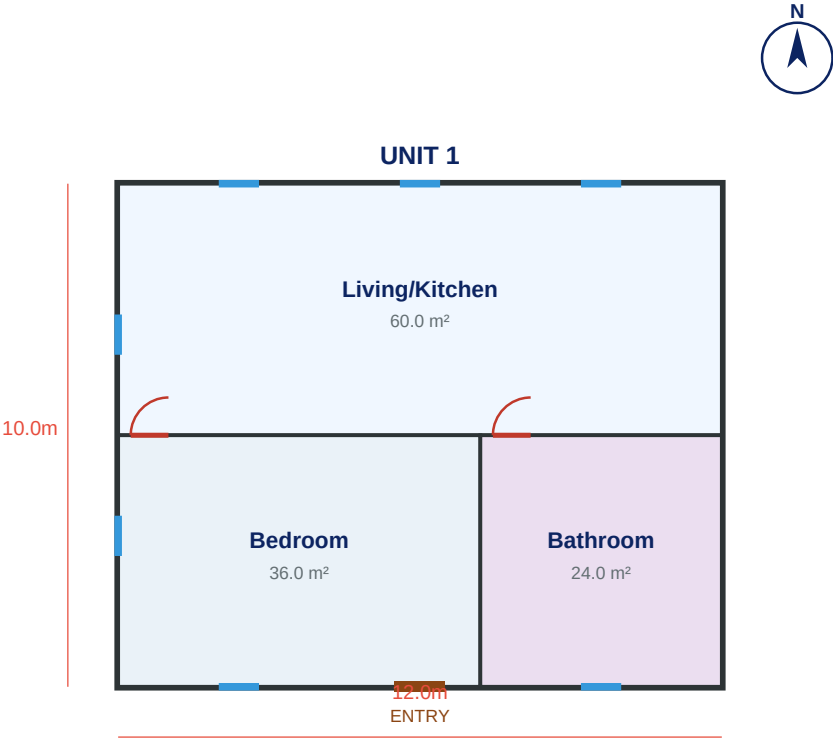
SITE PLAN - Nairobi, Kenya | Plot: 30m×25m



10. ARCHITECTURAL FLOOR PLAN

Ground floor layout — 1 unit(s), 1 bedrooms, showing room dimensions, doors, windows and circulation.

FLOOR PLAN - 100.0m² | 1 Units | 1 Storey(s)



EMERSON EIMS BUILDING SUITE PRO

Area: 100.0m² | 1 Units × 1 Bed

Floor Plan | Scale 1:50 | 2026-04-27 | Drawing: FP-001

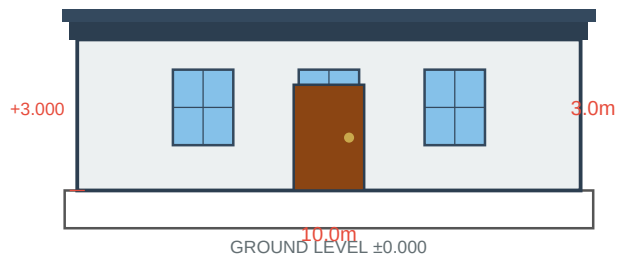
IFC Ready | BIM Level 2

11. BUILDING ELEVATIONS

4a. NORTH ELEVATION

Front view showing 1 storey(s), window/door openings, roof profile and floor levels.

NORTH ELEVATION - 1 Stories | Style: Modern



EMERSON EIMS | NORTH ELEVATION

Scale 1:50 | 2026-04-27 | Drawing: EL-N01

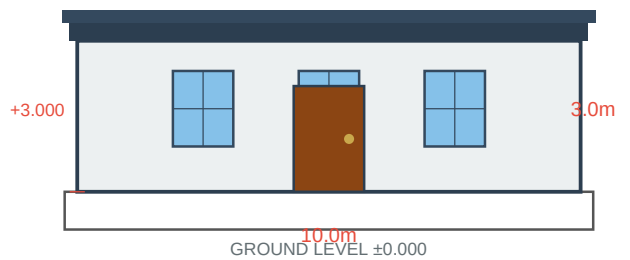
Stories: 1 | Height: 3.0m

Style: Modern | IFC Ready

4b. EAST ELEVATION

Side view showing building depth, roof slope and storey heights.

EAST ELEVATION - 1 Stories | Style: Modern



EMERSON EIMS | EAST ELEVATION

Scale 1:50 | 2026-04-27 | Drawing: EL-E01

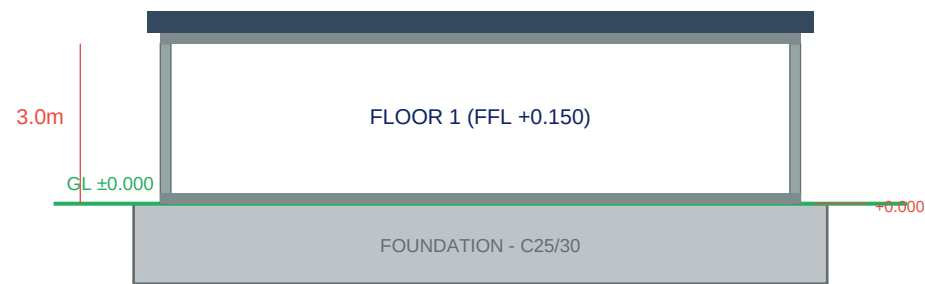
Stories: 1 | Height: 3.0m

Style: Modern | IFC Ready

12. BUILDING CROSS SECTION

Vertical cut through building showing floor slabs, storey heights (1 floors), foundation depth, roof structure and structural members.

CROSS SECTION A-A | 1 Stories



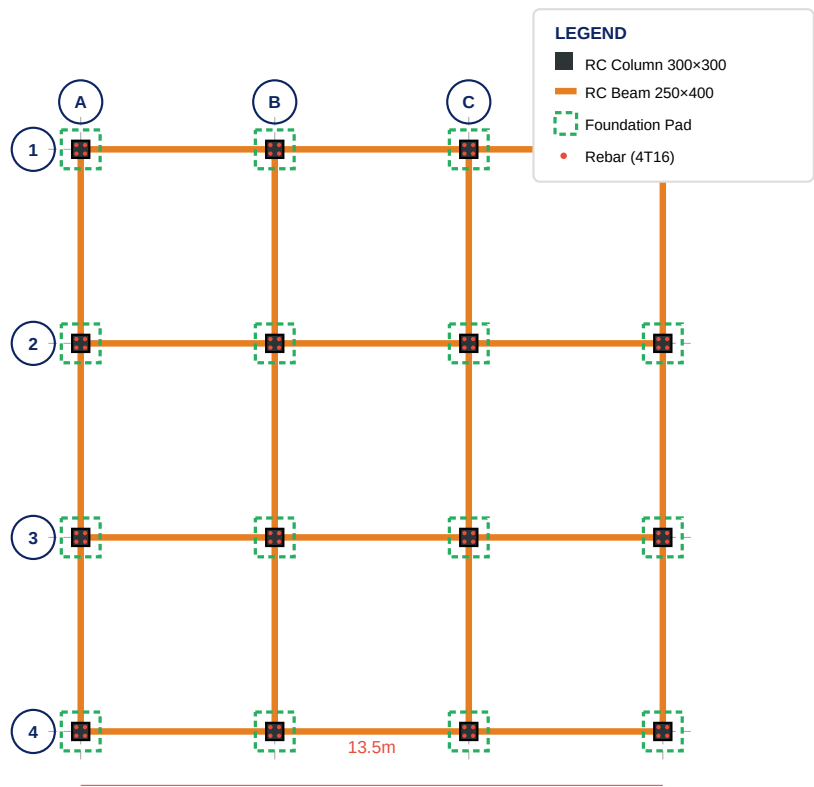
EMERSON EIMS | BUILDING SECTION A-A

2026-04-27 | SEC-001 | Scale 1:50

13. STRUCTURAL LAYOUT

Column grid, beam layout and foundation pads. Foundation type: Pad. All dimensions in metres. Concrete grade C25/30, Steel Y16.

STRUCTURAL LAYOUT - Pad Foundation | 1F | Grid 4.5m×4.5m



EMERSON EIMS | STRUCTURAL LAYOUT

Grid: 4.5m × 4.5m | Columns: 16 | 2026-04-27 | STR-001

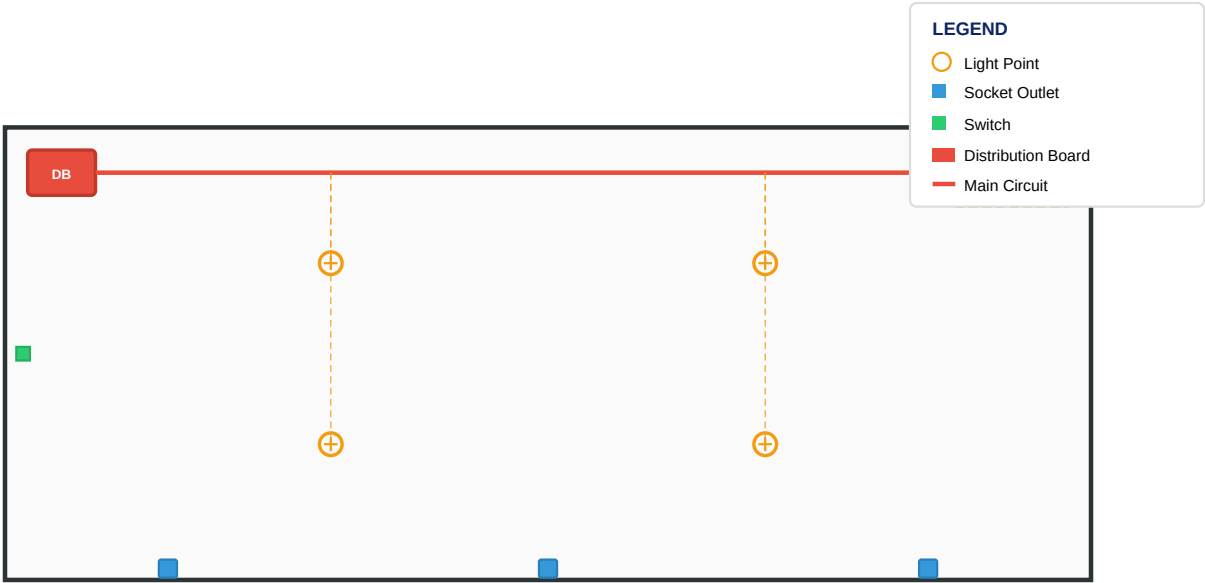
Concrete: C25/30 | Steel: Y16

Foundation: Pad | FOS ≥ 3.0

14. MEP LAYOUT — ELECTRICAL

Electrical distribution layout showing DB positions, circuit runs, socket/light points and earthing.

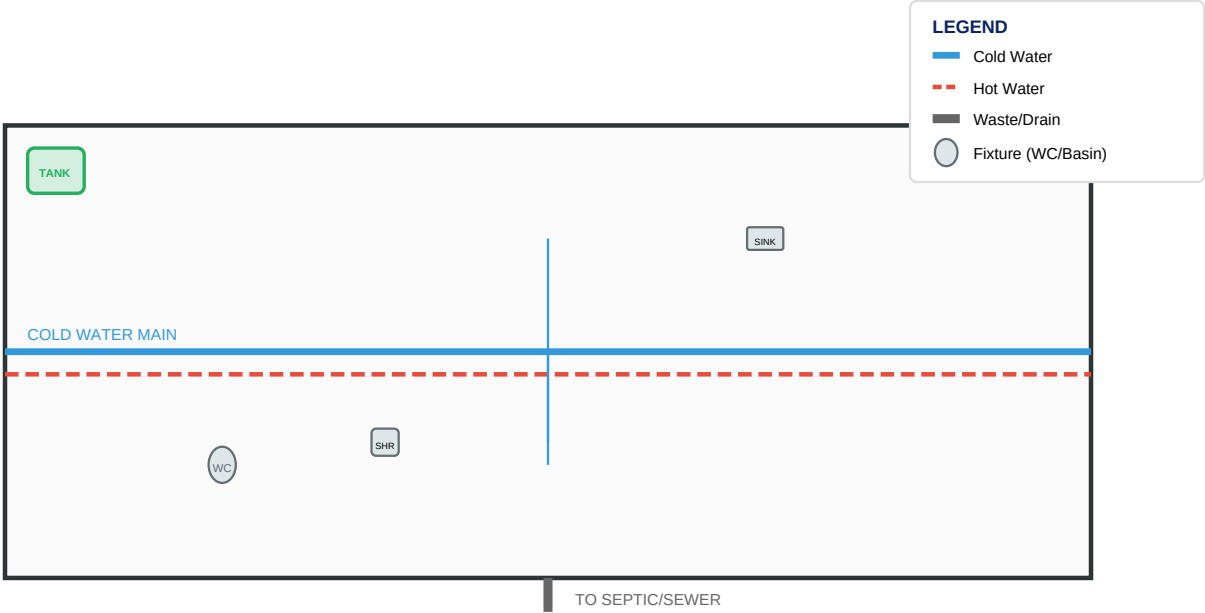
ELECTRICAL LAYOUT - 100.0m² | 1 Units



15. MEP LAYOUT — PLUMBING

Plumbing layout showing supply/drain runs, fixture locations, water heater and main shutoff valve.

PLUMBING LAYOUT - 100.0m² | 1 Units



16. COST SUMMARY

Item	Amount (KES)
Material Cost	KSh 4,944,715.10
Labor Cost	KSh 556,280.45
Equipment Cost	KSh 395,577.21
Overhead	KSh 440,079.64
Contingency	KSh 633,665.24
Subtotal	KSh 6,970,317.64
Contractor Profit	KSh 1,045,547.65
Total Project Cost	KSh 8,015,865.29
Cost Per Sqm	KSh 80,158.65
Labor Multiplier	KSh 0.25
Fx From Usd	KSh 130.00

This report was generated by EMERSON EIMS Building Suite Pro. All drawings are indicative and must be verified by a licensed engineer before construction.